

Credit Card Fraud Detection Using Machine Learning with Unbalanced Dataset

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Abstract: In this paper we focused on the problem of detecting credit card fraud activity in a timely manner. We worked with a dataset containing transactions from 2013-2014 that were labeled as either fraud or non-fraud activity. We used various machine learning models (Logistic Regression, Gaussian Naive Bayes, Decision Tree, Random Forrest) to analyze the data and try to identify patterns that might indicate fraudulent activity. Because the dataset was imbalanced, with a low number of fraud activity records, we generated synthetic fraud activity records in order to improve the performance of the models. We were able to achieve a improved experimental results with syntetically populated dataset.

1) Introduction:

Credit card fraud is the unauthorized use of a credit or debit card to make purchases, withdraw cash, or make other transactions. In recent years It can happen if a third-party uses customers credit-card or credit-card information without users authorization or permission. Credit card fraud can cause financial losses for both cardholders and banks. Users can take measures to protect themselves from credit card fraud, but it may not be enough. As for the Bank, they use advanced software to monitor credit card transactions for suspicious activity. If the software detects a potentially fraudulent transaction, the bank can take steps to prevent it from being completed. Many banks offer zero liability protection, which means that customers will not be held responsible for any unauthorized charges on your credit card. Banks can send fraud alerts to customers if they detect suspicious activity on the customer's account. This can help customers stay informed and take action to prevent further fraud. If customers credit card is stolen or compromised, the bank can issue you a new card with a new card number. Also, some banks offer credit monitoring services to their customers to help them keep track of their credit reports and detect any unusual activity. By

implementing these and other security measures, banks can help protect their customers from credit card fraud. Other than that customer spending patterns tend to vary and can change frequently, especially during holidays or on days when there are special discounts available. This makes it difficult to accurately predict and understand the behavior of customers and detect fraudulent activity. Fraudsters often take advantage of this ambiguity in spending patterns to conceal their fraudulent activities. Researchers are therefore interested in using a combination of unsupervised learning techniques (which do not require labeled training data) with supervised learning methods (which do require labeled training data) to develop classifiers for credit card fraud detection. By combining the strengths of both approaches, it may be possible to improve the accuracy and effectiveness of fraud detection systems. [1] For example "HOBA" stands for "homogeneity-oriented behavior analysis," and it is a feature engineering framework that is designed to "profile" users in order to detect fraudulent activity. The idea is that by creating profiles of users based on their normal behavior, it is possible to detect when a user or a fraudster deviates from their usual patterns or behaviors. This can be useful in identifying fraudulent activity, as it is often the case that fraudulent activity does not follow the same patterns as normal, legitimate activity. By analyzing the homogeneity or consistency of a user's behavior, it is possible to detect when there are deviations from the expected patterns, which can be a sign of fraudulent activity. [2] When working with imbalanced or skewed datasets, it is often necessary to try a variety of approaches in order to find the best fit. This can involve feature engineering, such as generating synthetic samples or reweighting the classes, as well as experimenting with different machine learning algorithms and their hyperparameters. It is important to keep in mind that no single approach will work for all datasets, and it may be necessary to try multiple methods in order to find the one that gives the best results. [13]

2) Related work:

According to the information provided, researchers have compared multiple machine learning models for the purpose of detecting fraudulent activity. They found that all of the models had an accuracy of approximately 0.99, but the time it took to achieve these results varied significantly, ranging from 20 seconds to 583 seconds. The researchers also noted that each machine learning model had its own advantages and disadvantages. For example, they mention that logistic regression (LR) is prone to overfitting, random forests (RF) are better suited to handling large amounts of data, and support vector machines (SVM) are not transparent. Despite the varying strengths and weaknesses of the different models, the researchers conclude that a deep convolutional neural network (DCNN) is the best option for fraud detection. [3] According to researchers in [4] the SVM algorithm had an accuracy of 0.9349, a precision of 0.9743, and a recall of 0.8976. The KNN algorithm had an accuracy of 0.9982, a precision of 0.7142, and a recall of 0.393. The ANN algorithm had an

accuracy of 0.9992, a precision of 0.8115, and a recall of 0.7619. Based on these results, the text concludes that the ANN algorithm is the best performing model of all detecting credit card fraud activity. The main problem that the researchers aim to address is that fraudulent activity represents a very small percentage of all credit card transactions, making it difficult to accurately detect. To tackle this problem, the researchers propose using an optimized light gradient boosting machine (LGBM) with a Bayesian-based hyperparameter optimization algorithm and they achieve 98% accuracy with this method.[5] Another approach is from researchers in [6] that using GAN method to populate fraudulent activity so that the dataset can be easier to learn for the model. Synthetic Minority Oversampling Technique (SMOTE) also used then models are compared and they both achieved 99% accuracy where the models almost did not differ in other metrics.

In [8] researchers compared the performances of LR, NB, RF, MLP models and found that Random Forrest had the best performance all around based on accuracy, precision and recall. In order to improve the accuracy of fraud detection on a highly imbalanced dataset, researchers [10] used a combination of multiple algorithms and employed a majority voting system. They also experimented with using AdaBoost to evaluate the results. The goal was to utilize multiple models, rather than just a single one, to make a decision about whether an activity was fraudulent. The motivation for this approach was that the dataset had a low number of fraud activity records, making it difficult for a single model to learn effectively. Long Short-Term Memory (LSTM) model and a Gated Recurrent Unit (GRU) model. Both of these models are types of Recurrent Neural Networks (RNNs), which are particularly well-suited for handling sequential data. It seems that the LSTM model had an accuracy of 0.95, while the GRU model significantly outperformed a baseline Artificial Neural Network (ANN) model, with an accuracy of 0.89. This suggests that the GRU model was more effective at making predictions on the dataset than either the LSTM model or the ANN model. [13]

3) Dataset:

Dataset used is the transactions made by customer in a European bank in the year 2013–14. It consist of 31 columns, in which 30 columns are the features and the one class is the target class which decides about whether the transaction is fraud or non valid. V1-V28 features are the results of PCA, which had to be anonymized to be published as publishers of this data set claims. Other features are time, amount and class. Class being fraud or not fraud encoded as 1 or 0. This data set contains **284.807 activity records in total**. This dataset contains **287.325 non-fraud activity and 492 fraudelent activity records**. As can be seen from these number, this data-set is highly skewed it is often said to be necessary to use oversampling or reweighting, in order to handle skewed datasets effectively. Type of parameters are float for 30 features and integer for Class feature.

4) Model Comparisons: In this research we tried different models with the regular dataset and with the populated fraud-activity with synthetic data to improve performance.

Non populated data results:

Model/Metric	Precision	Accuracy	Recall	F1-Score
Logistic Regression	0.7458	0.9990	0.6929	0.6527
Gaussian Naive Bayes	0.1463	0.9930	0.23	0.6489
Decision Tree	0.7387	0.9990	0.7413	0.7448
Random Forrest	0.9426	0.9995	0.8370	0.7533

According to experimental results Random Forrest achieved the best results overall.

Populated data: Populated data now has 5686430 transaction records. Dataset is now consists 50% fraud and 50% non-fraud activity records.

Logistic Regression	0.7926	0.9990	0.7126	0.6490
Gaussian Naive Bayes	0.1510	0.9928	0.2448	0.6491
Decision Tree	0.7588	0.9991	0.7599	0.7626
Random Forrest	0.9497	0.9995	0.8457	0.7628

Before and after populating data, Random Forrest was the best performer detecting fraud activity.

Discussion: Setting up a monitoring systems to track the model's performance and to alert you if it begins to degrade is a good idea. We may need to periodically retrain the model to incorporate new data and to maintain its performance.

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Link to dataset:

Credit Card Fraud Detection

Anonymized credit card transactions labeled as fraudulent or genuine, MACHINE LEARNING
GROUP – ULB, [Credit Card Fraud Detection | Kaggle](#)